

Business Requirements Document (BRD)

University Library Management System

Version	1.1
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1. Executive Summary

The University Library Management System is a digital solution designed to automate student borrowing activities. The system aims to provide real time visibility into book availability, streamline the reservation process, and manage loan renewals based on specific business rules.

2. Business Scenario

The library requires a system that manages high volume transactions while enforcing borrowing limits. The system must act as a real time interface between the student and the library's central database to prevent double bookings and manage overdue risks.

3. Functional Requirements

ID	Requirement Description	Priority
FR-01	User Authentication: Students must authenticate using a valid Student ID and Password. The system shall display specific error messages for failed attempts.	Critical
FR-02	Online Catalog Search: Users shall be able to search for books and view real time availability, author, and shelf location.	High
FR-03	Book Detail View: Students shall be able to view detailed information (metadata) about a selected book before	Medium

	placing a reservation.	
FR-04	Reservation Management: The system must validate that the user has not exceeded the maximum reservation limit before confirming a book hold.	High
FR-05	Conditional Renewal: The system shall allow book renewals ONLY if there are no active reservations for that specific book by other students.	Critical
FR-06	Loan Reminders: The system shall automatically extend the borrowed book check process to send due date reminders to students.	Low
FR-07	Real time Database Sync: The system must update the Library DBMS immediately after every search, reservation, or renewal action.	Critical

4. Stakeholder Roles (Actors)

- **Student / Customer:** Searches for books, places reservations, and manages renewals through the self service portal.
- **Library DBMS:** The central database storing real-time status of books and user borrowing history.

5. Business Rules & Logic

- **Renewal Logic:** A renewal is successful ONLY IF (Book Status = Borrowed) AND (Reservation Count = 0).
- **Reservation Logic:** A reservation is successful ONLY IF (Book Status = Available) AND (User Current Reservations < Max Limit).
- **Data Integrity:** Every action that modifies book status must be synchronized with the DBMS in real-time to prevent double bookings.